

LABORATORIO 7

PROBLEMA 2:

1. $S \rightarrow 0AO/1B1/BB$ a) $S \rightarrow 0AO/1B1/BB/00/11/B/\epsilon$

$A \rightarrow C$

$A \rightarrow C$

$B \rightarrow S/A$

$B \rightarrow S/A$

$C \rightarrow S/\epsilon$

$C \rightarrow S$

b) $S \rightarrow B \rightarrow A \rightarrow C \rightarrow S$ $S \rightarrow 0AO/1B1/BB/00/11/\epsilon$

$A \rightarrow C \rightarrow S \rightarrow B \rightarrow S$ $A \rightarrow 0AO/1B1/BB/00/11$

$B \rightarrow A \rightarrow C \rightarrow S$ $B \rightarrow 0AO/1B1/BB/00/11$

$C \rightarrow S \rightarrow A \rightarrow B \rightarrow C$ $C \rightarrow 0AO/1B1/BB/00/11$

c) $\{S, A, B, C\} \rightarrow 00$ $S \rightarrow A$ $S \rightarrow 0AO/1B1/BB/00/11/\epsilon$

$\{S, A, B, C\} \rightarrow 11$ $S \rightarrow B$ $A \rightarrow 0AO/1B1/BB/00/11$

$C \rightarrow \text{no alcanzable}$ $B \rightarrow 0AO/1B1/BB/00/11$

d) $S \rightarrow z_0 y_0 / z_1 y_1 / z_0 z_0 / z_1 z_1 / BB / \epsilon$

$A \rightarrow z_0 y_0 / z_1 y_1 / z_0 z_0 / z_1 z_1 / BB$

$B \rightarrow z_0 y_0 / z_1 y_1 / z_0 z_0 / z_1 z_1 / BB$

$y_0 \rightarrow A z_0$

$y_1 \rightarrow B z_1$

$z_0 \rightarrow 0$

$z_1 \rightarrow 1$

2. $S \rightarrow aAa \mid bBb \mid \epsilon$ a). $S \rightarrow aAa \mid aa \mid bBb \mid bb \mid \epsilon$

$A \rightarrow C \mid a$

$A \rightarrow C \mid a$

$B \rightarrow C \mid b$

$B \rightarrow C \mid b$

$C \rightarrow CDE \mid \epsilon$

$C \rightarrow CDE \mid DE$

$D \rightarrow A \mid B \mid ab$

$D \rightarrow A \mid B \mid ab$

b). $A \rightarrow C$ $S \rightarrow aAa \mid aa \mid bBb \mid bb \mid \epsilon$

$B \rightarrow C$

$A \rightarrow CDE \mid DE \mid a$

$D \rightarrow A$

$B \rightarrow CDE \mid DE \mid b$

$D \rightarrow B$

$C \rightarrow CDE \mid DE \mid$

$D \rightarrow CDE \mid DE \mid a \mid b \mid bb$

c). $S \rightarrow aa \mid bb$

$A \rightarrow a$

$S \rightarrow aAa \mid aa \mid bBb \mid bb \mid \epsilon$

$A \rightarrow a$

$B \rightarrow b$

$B \rightarrow b$

$D \rightarrow a \mid b \mid ab$

$D \rightarrow a \mid b \mid ab$

$S \rightarrow aAa \mid aa \mid bBb \mid bb \mid \epsilon$

$A \rightarrow a$

$B \rightarrow b$

d). $S \rightarrow ZaYa \mid ZbYb \mid ZaZa \mid ZbZb \mid \epsilon$

$A \rightarrow a$

$B \rightarrow b$

$Y_a \rightarrow AZa$

$Y_b \rightarrow BZb$

$Za \rightarrow a$

$Zb \rightarrow b$

3. $S \rightarrow ASA / aB$	a) $S \rightarrow ASA / SA / AA / AS / A / S / aB / a / \epsilon$
$A \rightarrow B / S$	$A \rightarrow B / S$
$B \rightarrow b / \epsilon$	$B \rightarrow b$

b) $S \rightarrow A$	$S \rightarrow ASA / SA / AA / AS / aB / a / b / \epsilon$
$S \rightarrow S$	$A \rightarrow ASA / SA / AA / AS / aB / a / b$
$A \rightarrow B$	$B \rightarrow b$
$A \rightarrow S$	

c) $B \rightarrow b$	$S \rightarrow ASA / SA / AA / AS / aB / a / b / \epsilon$
$A \rightarrow b / a$	$A \rightarrow ASA / SA / AA / AS / aB / a / b$
$S \rightarrow a / b / \epsilon$	$B \rightarrow b$

d) $S_0 \Rightarrow AX / SA / AA / AS / \epsilon aB / a / b / \epsilon$
$S \Rightarrow AX / SA / AA / AS / \epsilon aB / a / b$
$A \Rightarrow AX / SA / AA / AS / \epsilon aB / a / b$
$B \rightarrow b$
$X \rightarrow SA$
$\epsilon a \rightarrow a$